

Homework #4

CS5402 — Intro To Data Mining

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1 Support Vector Machine

1. Our equations would be as follows:

$$\begin{aligned}\alpha_1 \varphi(s_1) \cdot \varphi(s_1) + \alpha_2 \varphi(s_2) \cdot \varphi(s_1) &= 1 \\ \alpha_1 \varphi(s_1) \cdot \varphi(s_2) + \alpha_2 \varphi(s_2) \cdot \varphi(s_2) &= -1\end{aligned}$$

2. Assuming φ to be the identity operator, we get as follows:

$$\begin{aligned}6\alpha_1 + 11\alpha_2 &= 1 \\ 11\alpha_1 + 21\alpha_2 &= -1\end{aligned}$$

From this, we get as follows:

$$\alpha_1 = 32/5 \quad \alpha_2 = -17/5$$

3. Our discriminating hyperplane would be as follows:

$$w' = \langle -4/5, -2/5, 3 \rangle$$

Therefore, the plane equation would be as follows:

$$-4/5 x - 2/5 y + 3 = 0$$

2 K2 Bayesian Network

For our α values, we get the following:

$$\begin{aligned}\alpha_{411} &= 1 & (\# \text{ cases with } x_1 = 0, x_2 = N, x_4 = H) \\ \alpha_{412} &= 2 & (\# \text{ cases with } x_1 = 0, x_2 = N, x_4 = L) \\ \alpha_{413} &= 1 & (\# \text{ cases with } x_1 = 0, x_2 = N, x_4 = M) \\ \alpha_{421} &= 0 & (\# \text{ cases with } x_1 = 0, x_2 = Y, x_4 = H) \\ \alpha_{422} &= 3 & (\# \text{ cases with } x_1 = 0, x_2 = Y, x_4 = L) \\ \alpha_{423} &= 1 & (\# \text{ cases with } x_1 = 0, x_2 = Y, x_4 = M) \\ \alpha_{431} &= 1 & (\# \text{ cases with } x_1 = 1, x_2 = N, x_4 = H) \\ \alpha_{432} &= 3 & (\# \text{ cases with } x_1 = 1, x_2 = N, x_4 = L) \\ \alpha_{433} &= 1 & (\# \text{ cases with } x_1 = 1, x_2 = N, x_4 = M) \\ \alpha_{441} &= 0 & (\# \text{ cases with } x_1 = 1, x_2 = Y, x_4 = H) \\ \alpha_{442} &= 2 & (\# \text{ cases with } x_1 = 1, x_2 = Y, x_4 = L) \\ \alpha_{443} &= 1 & (\# \text{ cases with } x_1 = 1, x_2 = Y, x_4 = M)\end{aligned}$$

For our N values, we get the following:

$$\begin{aligned}N_{41} &= \alpha_{411} + \alpha_{412} + \alpha_{413} = 1 + 2 + 1 = 4 \\ N_{42} &= \alpha_{421} + \alpha_{422} + \alpha_{423} = 0 + 3 + 1 = 4 \\ N_{43} &= \alpha_{431} + \alpha_{432} + \alpha_{433} = 1 + 3 + 1 = 5 \\ N_{44} &= \alpha_{441} + \alpha_{442} + \alpha_{443} = 0 + 2 + 1 = 3\end{aligned}$$

3 Weka Output

Missing: assets/example-3

For the Naive vs Regular Bayesian Network, the results are similar. With the same confusion matrix, they both correct and correctly classify the same number of instances (and the same instances).

4 K2

From the graph, we can first determine that x_5 is not dependent on x_1, x_2, x_3 or x_4 . However, x_3 and x_4 depend on x_2 (from the arrows being directed from x_2 to x_3 and x_4). Likewise, x_2 depends on x_1 .